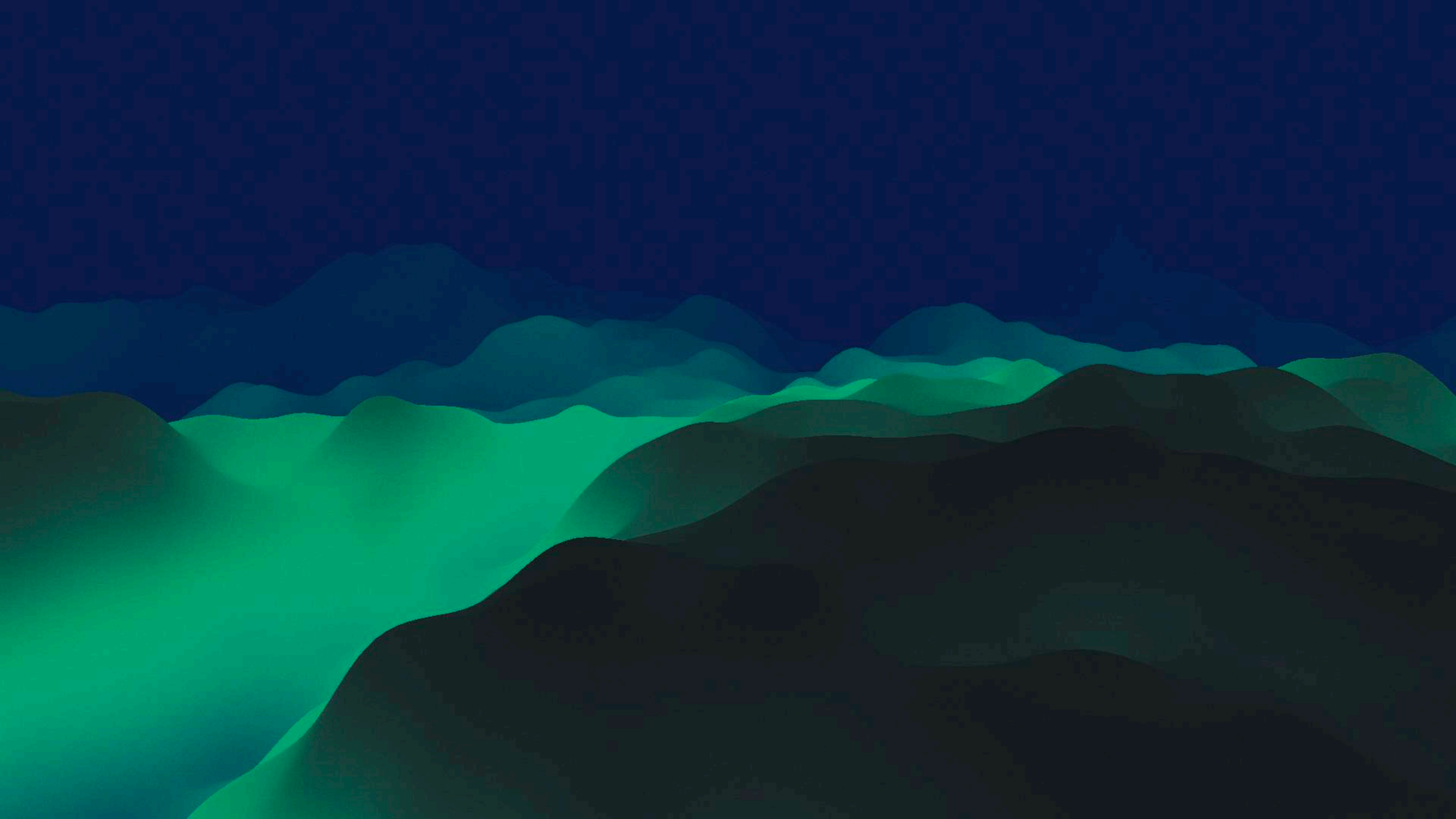






Redefining Hiring with Autonomous Intelligence

Multi-Agent HR Automation

Team: Rayyan, Deepesh, Jewel, Vaibhav

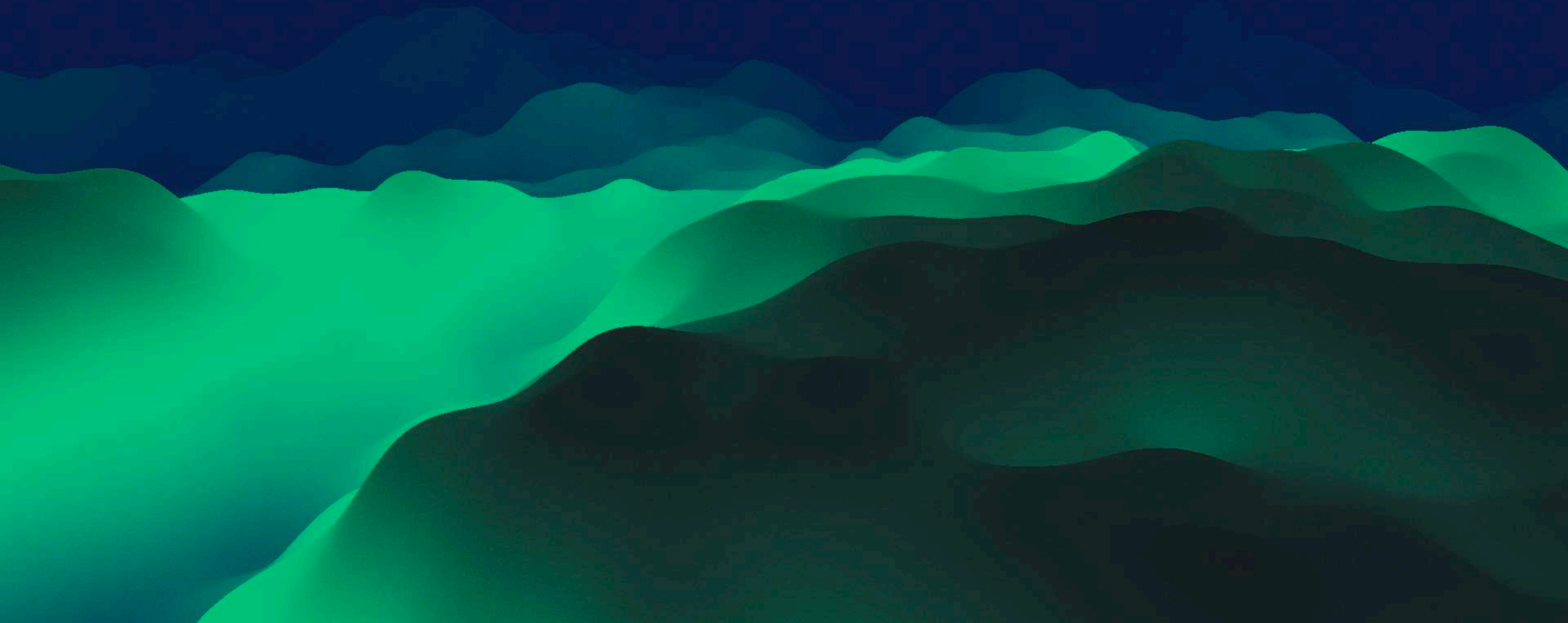
Mentor: Dheeraj

Module: Agentic AI | 2025

The Motivation & Problem Statement

Hiring is broken — inefficient, slow, and expensive.

Market & Opportunity



HireAI - Solution

Autonomous HR System with Multi-Agent AI

Flow of candidate



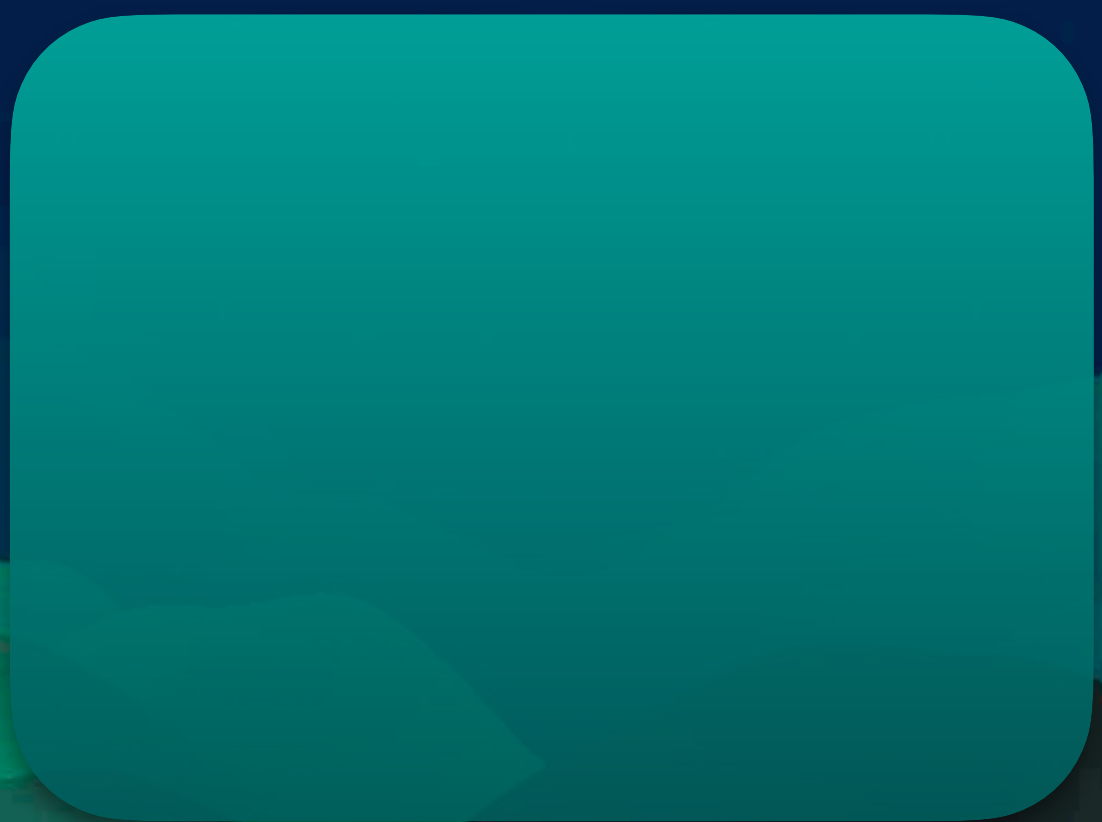
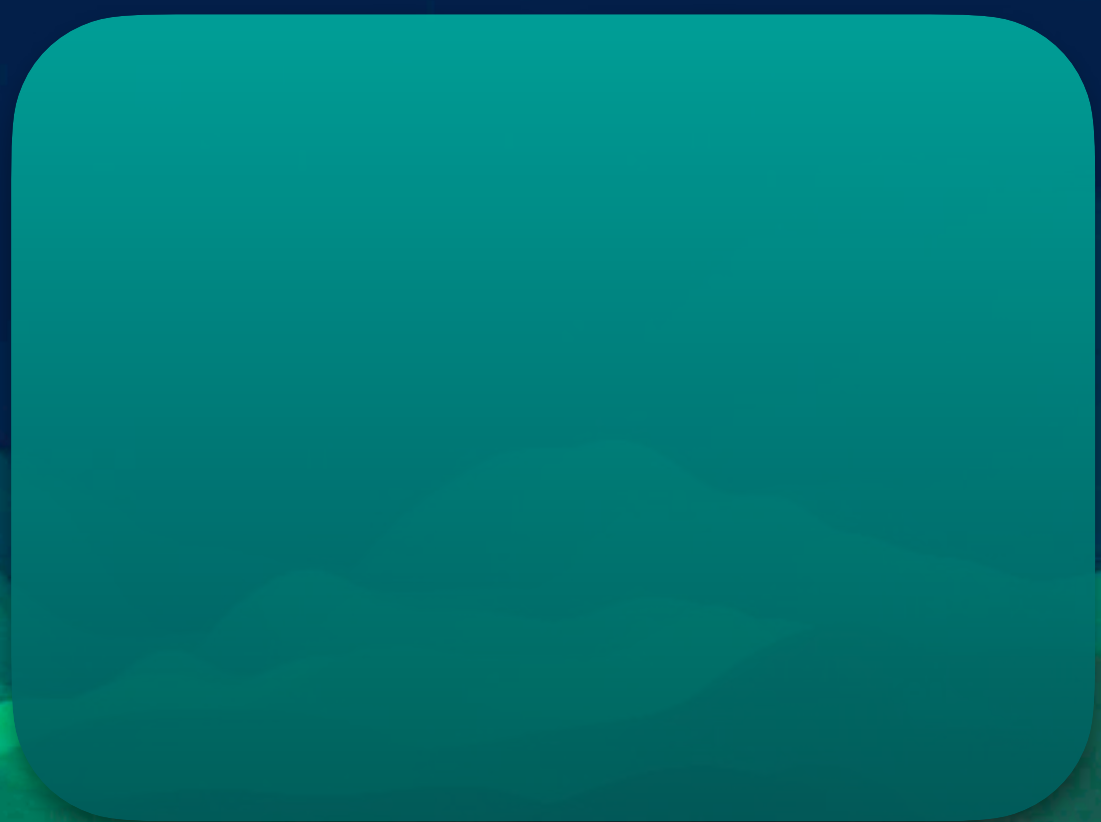
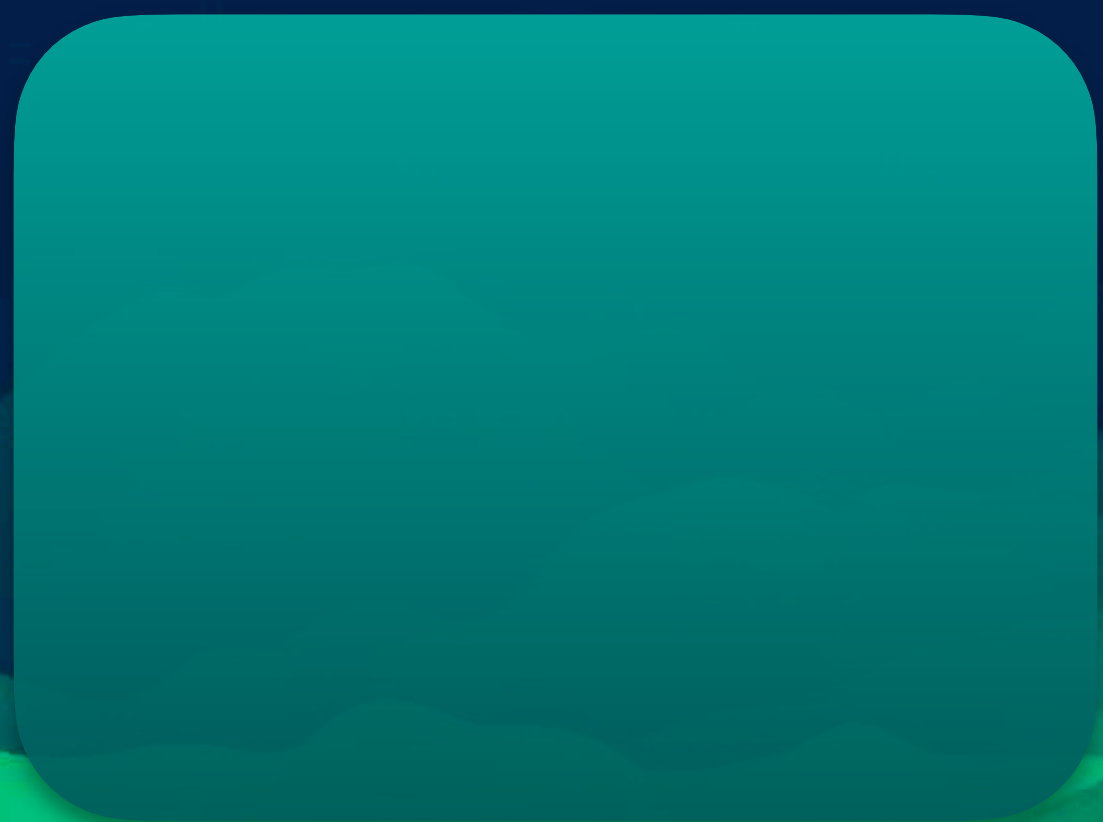
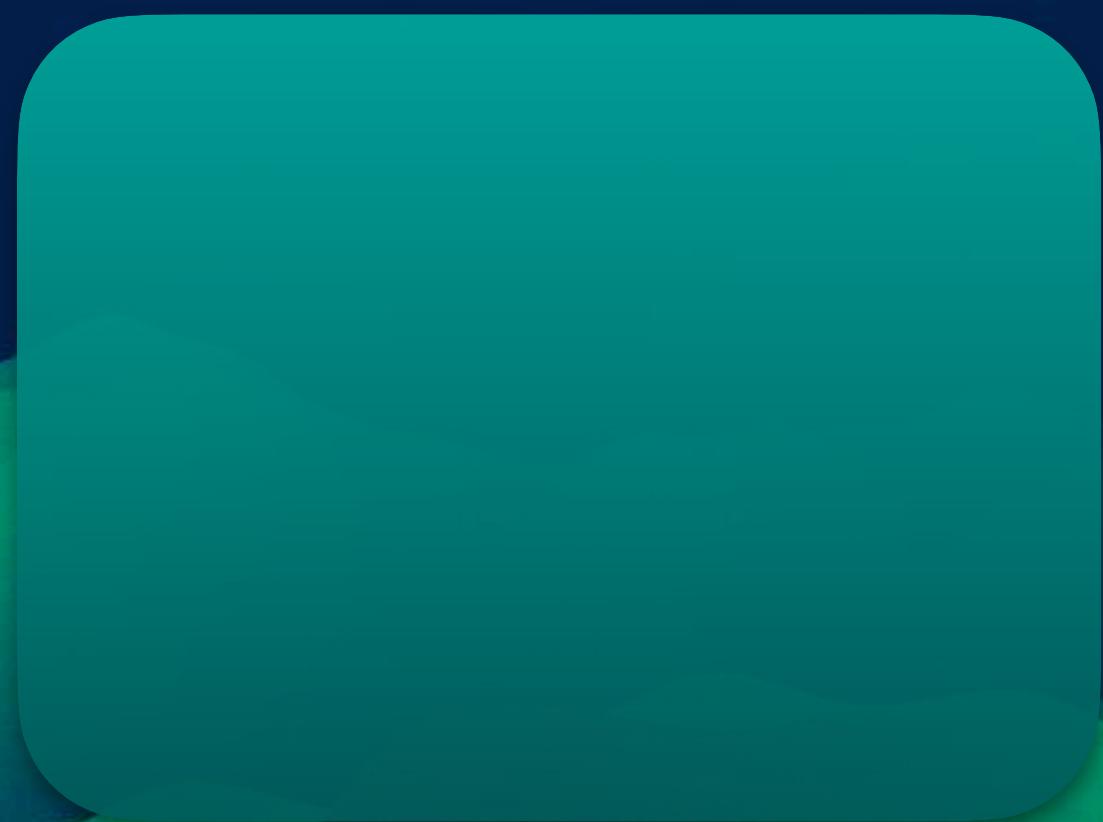
AI agents



decision report.

Project Objectives

- Automate resume screening, shortlisting, and interviews using **AI Agents**.
- Implement **CrewAI + LangChain** for multi-agent orchestration.
- Implement Guardrails for Fairness and Compliance.
- Use **LLMs, speech APIs, and embeddings** for holistic candidate evaluation.
- Build an **HR dashboard** to monitor candidates and insights.
- Reduce screening time and hiring cost while improving match accuracy.



Tech Stack

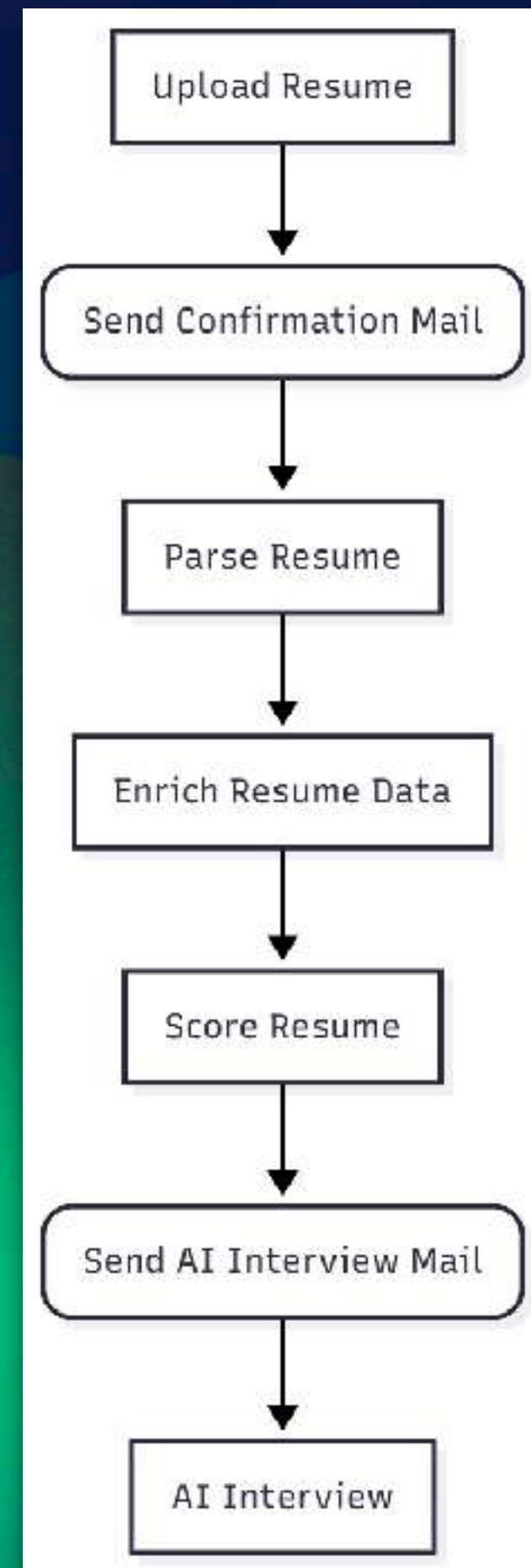
Layer	Key Technologies & Frameworks	APIs & Specific Tools
Frontend/Client	React/JavaScript Web App	WebSocket, Google Form + Apps Script
Backend/ Core Logic	FastAPI (High-Performance API), CrewAI (Workflow Orchestration), LangChain	REST API Endpoints: /forms/submit-application, /upload-resume/, /interview/start
AI/LLM Engine	Groq LLM, OpenAI LLM, Whisper (Speech-to-Text)	Speech Synthesis API (Text-to-Speech)
Data & Storage	MongoDB (Structured Data), FAISS Vector DB (Semantic Embeddings)	Google Drive (Temporary Resume Storage)
Integrations & Tools	MCP (Model Context Protocol), EmailTool (SMTP/SendGrid)	DatabaseTool (CRUD Ops), EmbeddingTool, ResumeParserTool

Macro Architecture Flow

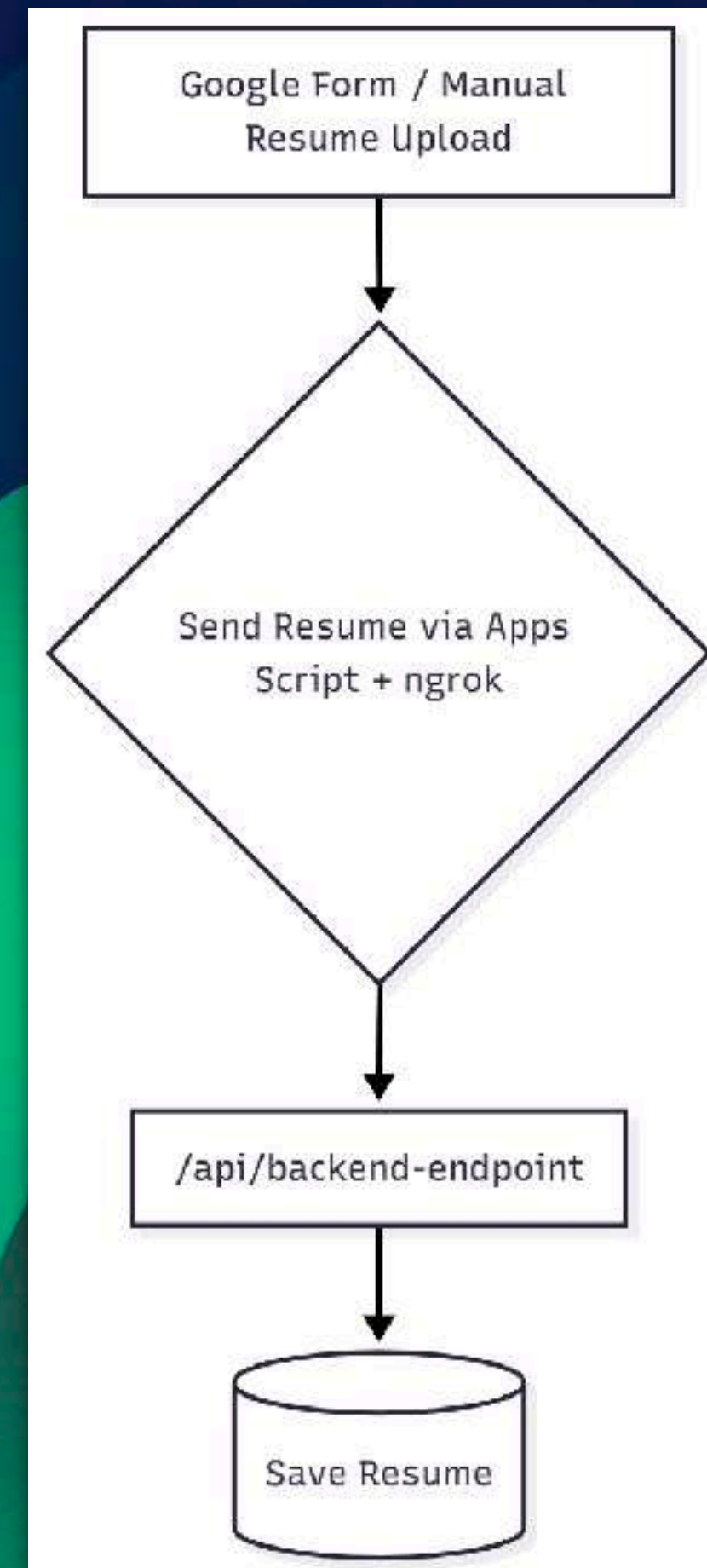
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System Architecture Overview

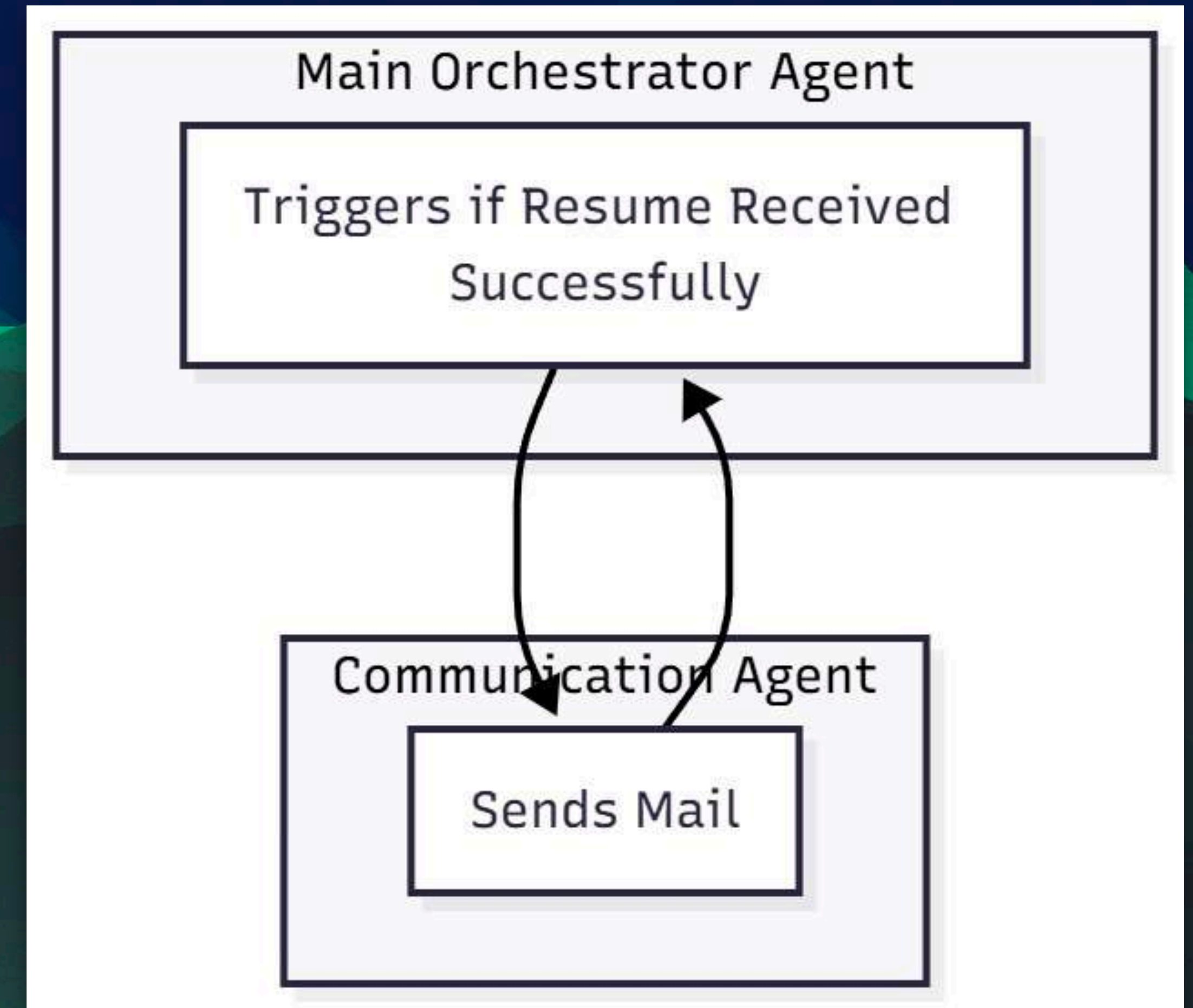
Micro Architecture Flow - 1



Overview tech flow chart



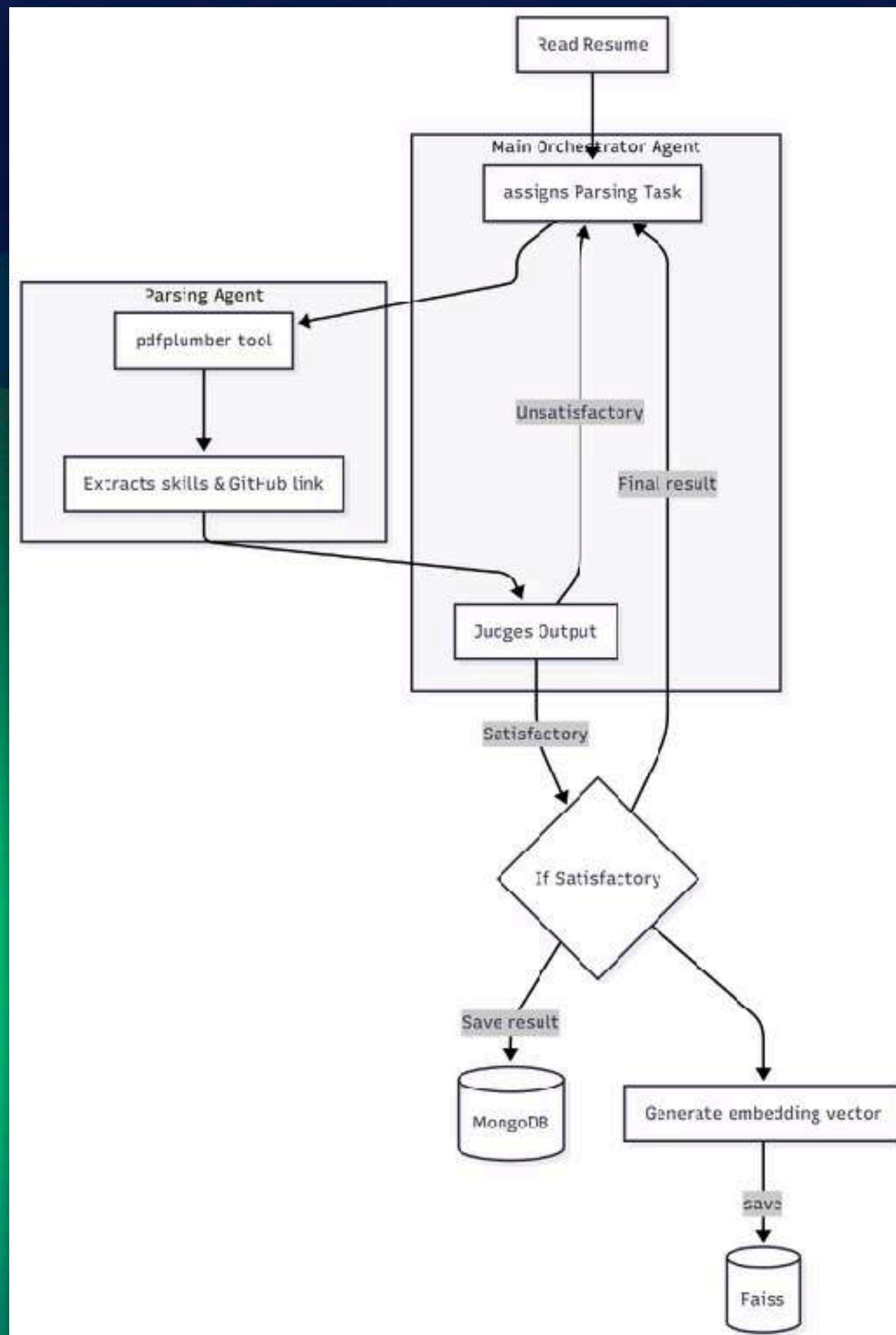
Upload resume flow chart



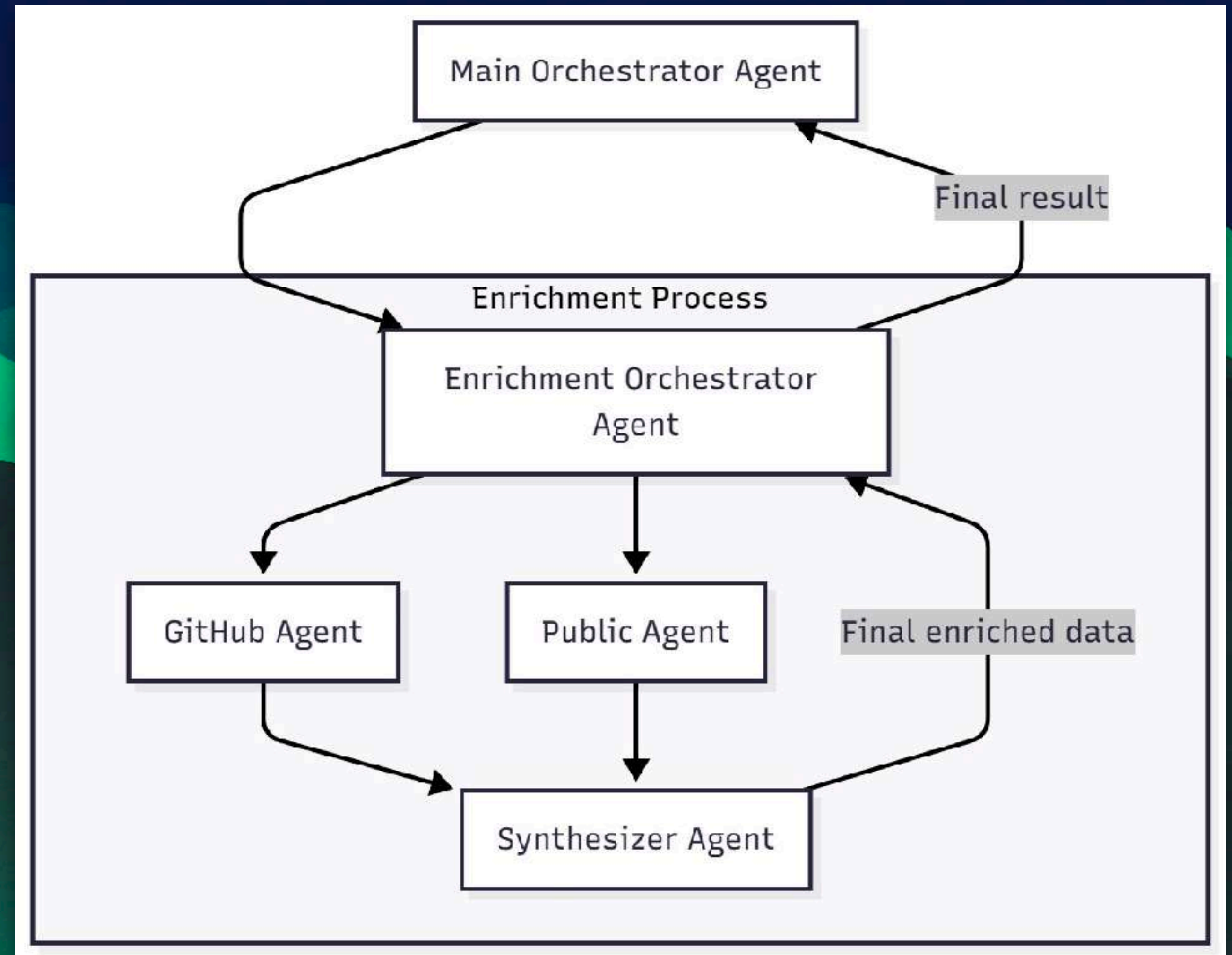
Send received mail flowchart

System Architecture Overview

Micro Architecture Flow - 2



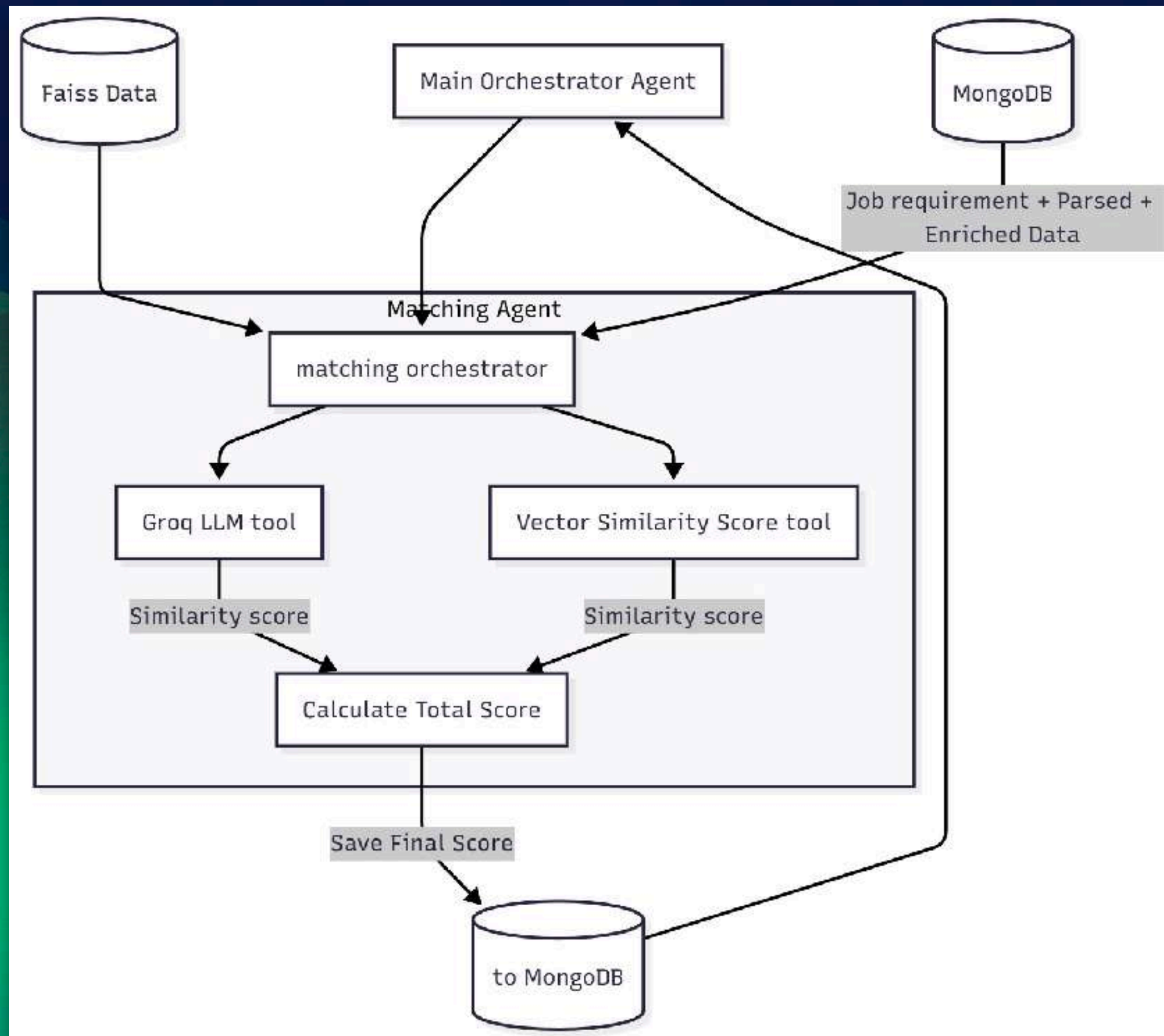
Parse resume flow chart



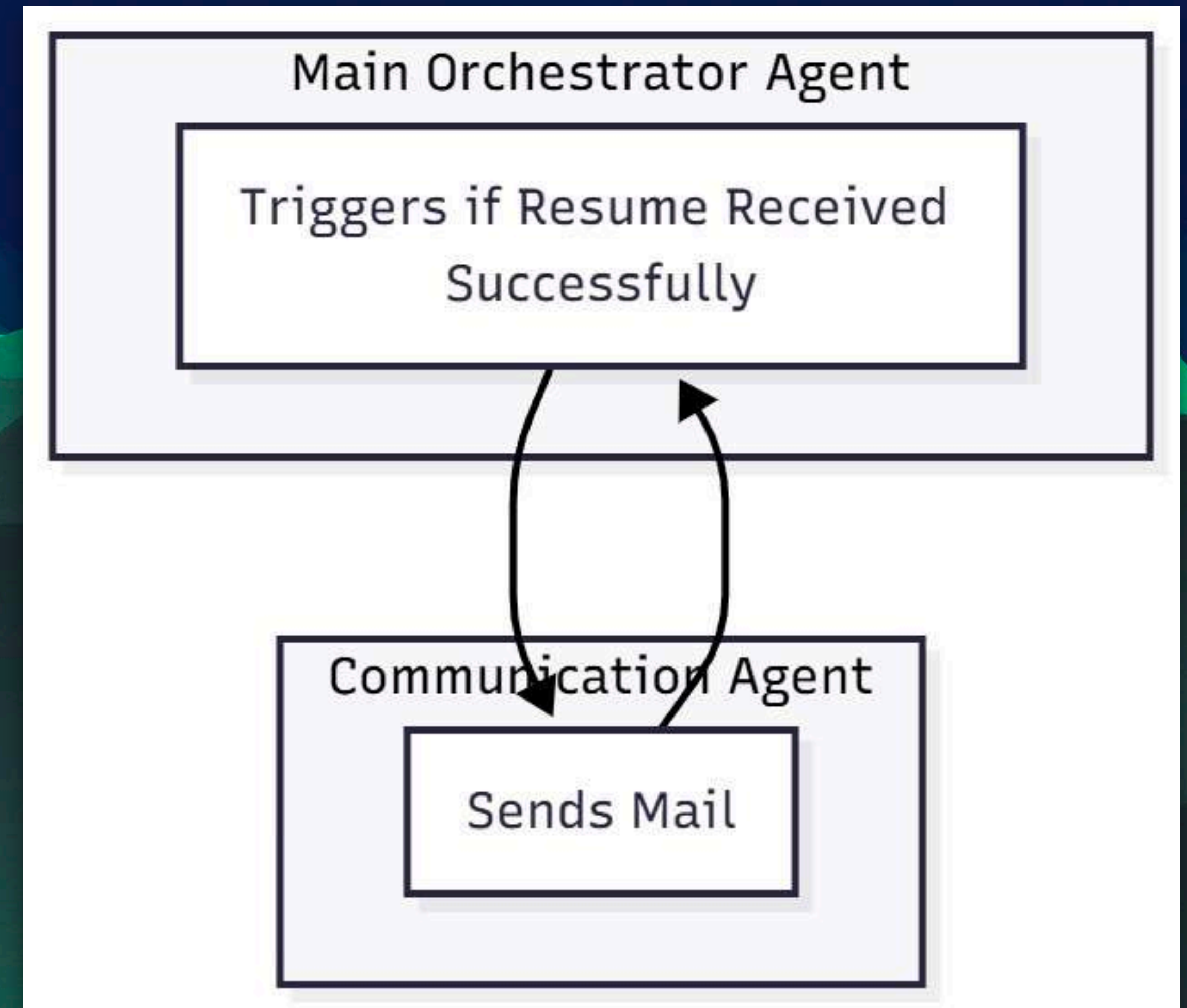
Enrich resume flow chart

System Architecture Overview

Micro Architecture Flow - 3



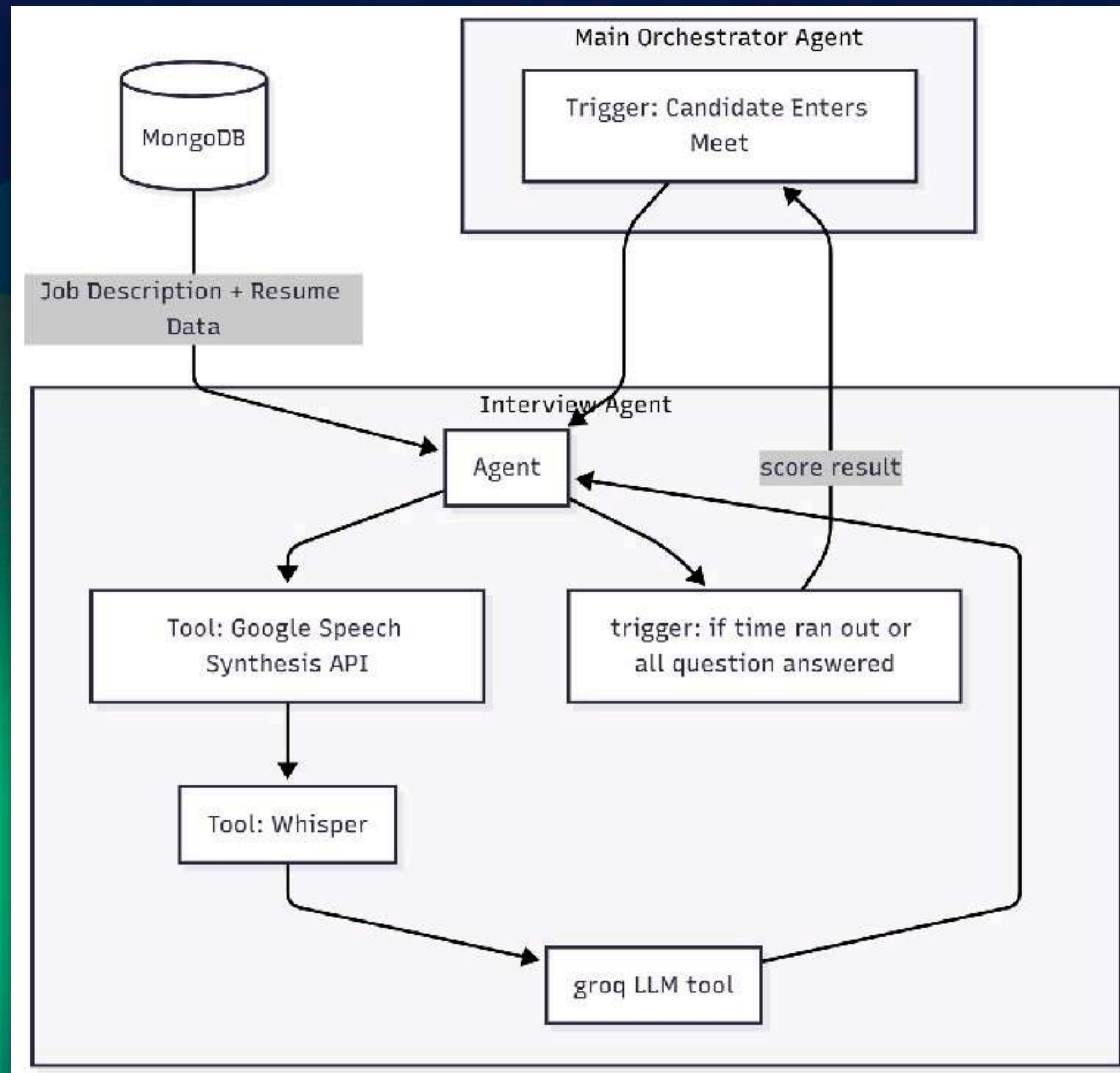
Scoring resume flowchart



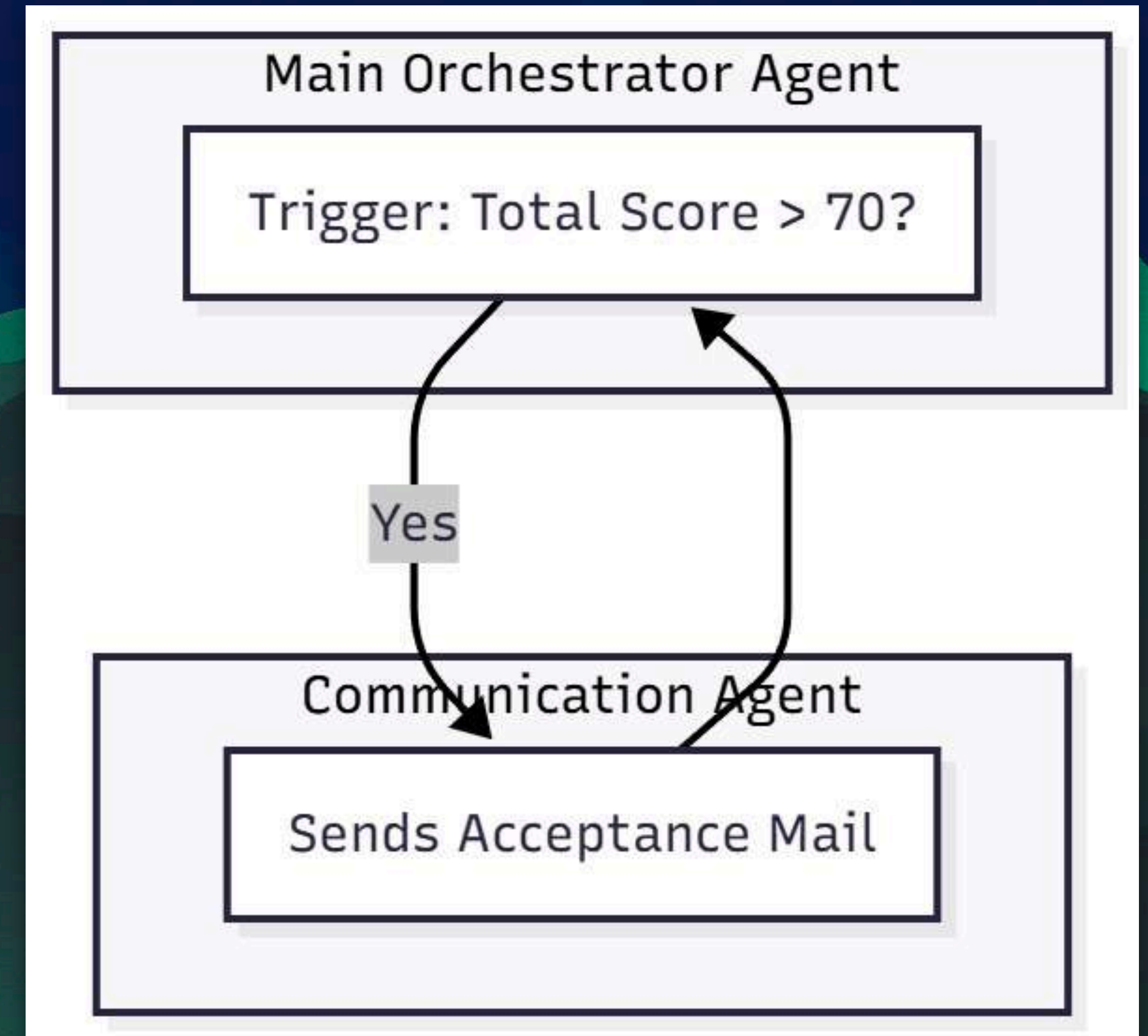
Send shortlisted mail flowchart

System Architecture Overview

Micro Architecture Flow - 4



AI interview



Send final acceptance mail flowchart

Project Outcomes

Metric	Before	After (Our System)
Screening Time	3–4 days	3 hours
Cost per Hire	₹8K–₹12K	₹3K–₹5K
Drop-off Rate	25–30%	12%
Accuracy	Manual review	85–90% fit precision

Risks & Mitigations

Risk	Mitigation Approach
Data Privacy & Security	Use access controls, encryption, minimal PII exposure
Model Hallucination / Error	Validation layers, fallback checks, human review
Agent Coordination Failures	Circuit-breaker logic, monitoring, retries
Integration / Adoption by HR Teams	Training, easy-to-use dashboard, onboarding process

Team & Roles

Member	Role & Responsibilities
Deepesh	Technical Lead – AI Systems & Orchestration Developed the Voice & Interview Agent , handled LLM and model integration , optimized agentic orchestration logic , and led the technical architecture implementation (FastAPI, CrewAI, MCP). Guided end-to-end system validation and deployment .
Rayyan Pasha	Research & System Design Led the problem research, validation, and conceptualization of the entire HR automation system. Defined system flow architecture , coordinated team roles, and built the pitch deck & narrative for technical + business coherence.
Jewel	Data Layer & Embedding Engineer Built and optimized vector storage & retrieval pipelines (FAISS + MongoDB) , handled resume vetting , and contributed to data preparation and candidate ranking mechanisms . Co-presenter for technical demo.
Vaibhav	UI & Communication Agent Engineer Developed the Dashboard UI , implemented the communication agent , managed API integrations between front-end and backend layers, and contributed to the technical demo voice + UX testing .

Future Directions & Extensions



Live Demo — System in Action

The background features a series of overlapping, wavy, horizontal bands in shades of green and blue. The top portion is a dark, deep blue, while the lower portion transitions into various shades of green, from a vibrant lime green on the left to a dark, forest green on the right. The waves create a sense of depth and movement, resembling a stylized landscape or a digital data visualization.

Takeaway / Learning Outcomes

- Gained practical understanding of **Agentic AI orchestration**.
- Integrated LLMs, Whisper, and FAISS into a real HR system.
- Learned data pipeline + embedding optimization.
- Built end-to-end workflow from **Google Forms → Dashboard**.
- Validated real-world HR inefficiencies through research.
- Strengthened teamwork and agile project execution skills.

Conclusion & Call to Action

- The HR / AI market is growing — the opportunity is strong
- Our system aims to bring automation, fairness, scalability to recruitment
- Demo showed working flow;
- Next step: pilot with real company
- We invite partners or users for deployment and feedback

References

1. Grand View Research — Global AI in HR market 2023 to 2030 [Grand View Research](#)
2. Reuters / Nasscom-BCG — India AI market to reach USD 17B by 2027 [NASSCOM, Reuters , India AI](#)
3. Grand View Research — HR software global market size 2023 to 2030 [Grand View Research](#)
4. PR Newswire — HR analytics market 2024 to 2031 [PR Newswire](#)
5. Nasscom AI Adoption Index / AI growth statements [NASSCOM](#)

The background features a series of overlapping, wavy, organic shapes in various shades of green and blue. The top portion is a solid dark blue, while the lower portion is filled with layers of green, ranging from a vibrant lime green to a deep forest green. The overall effect is a modern, digital landscape.

**“The best recruiters won’t be replaced by AI —
they’ll be augmented by it.”**